

Corporate Finance

Lecture 4: CAPM

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Plan

- ▶ Recap from lecture 3
- ▶ Ask what return a stock must offer to be worth adding to a portfolio: the required return and the beta
- ▶ Meet the CAPM: when every investor holds the tangency portfolio, that portfolio is the market itself
- ▶ Use the CAPM to determine the risk premium of any stock

Mostly based on the BDM textbook Chap. 11.6-8.

Recap lecture 3

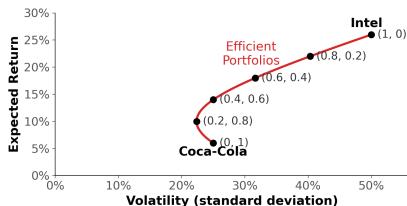
Recap of Lecture 3

Let's do a recap:

- ▶ Combining two stocks: many portfolios, many risk-return pairs
- ▶ Efficient portfolios and the efficient frontier
- ▶ The shape of the frontier depends on the correlation
- ▶ The minimum-variance portfolio
- ▶ Short sales extend the frontier
- ▶ From two stocks to N : the frontier moves left
- ▶ Adding the risk-free asset
- ▶ Borrowing and leverage
- ▶ The Sharpe ratio, and how it picks the efficient frontier
- ▶ The super-efficient (tangency) portfolio
- ▶ Two types of risk

Recap: combining two stocks

- ▶ Given two stocks, we choose the **weights** x_1 and x_2 , with $x_1 + x_2 = 1$. Each choice is one portfolio, with its own expected return and its own risk.
- ▶ Plotted as points, risk across and return up, all the mixes of the two stocks trace a **curve** (Intel and Coca-Cola, zero correlation):



Recap: efficient portfolios and the frontier

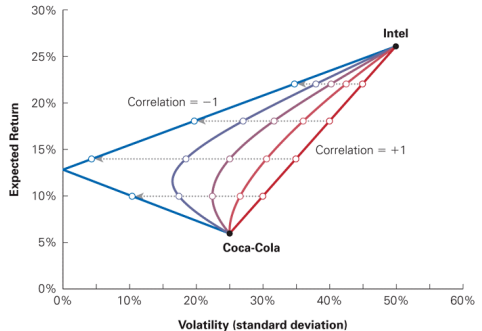
A portfolio is **efficient** when, for its level of risk, no other portfolio gives a higher return; equivalently, for its level of return, no other portfolio has a lower risk.

The **efficient frontier** is the set of all efficient portfolios: the upper part of the curve.

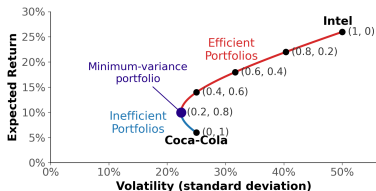
- ▶ Example: $(0, 1)$ and $(0.4, 0.6)$ both have 25% volatility, but the second returns 14% against 6%. Holding only Coca-Cola is dominated; the lower part of the curve is inefficient.

Recap: the shape depends on correlation

- ▶ **Correlation +1:** no diversification. The curve is a straight line between the two stocks.
- ▶ **Lower correlation:** the curve bends to the left, the same returns at lower risk.
- ▶ **Correlation -1:** the risks can cancel completely; with the right weights the portfolio has zero risk.



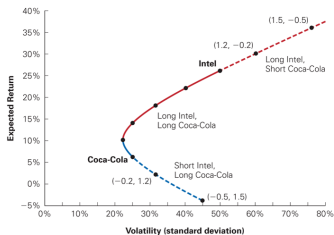
Recap: the minimum-variance portfolio



- ▶ The mix with the **lowest risk** of all: the “nose” of the curve, where the efficient set begins. Here (0.2, 0.8), with 22.4% volatility, below either stock alone.
- ▶ Its weights depend only on the volatilities and the correlation, **not on the expected returns**.

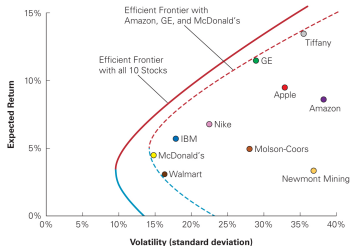
Recap: short sales extend the frontier

- ▶ A **short sale** is a negative weight: sell a stock you borrowed, and invest more than your own money in the other one.
- ▶ Shorting assets allows to extend the efficient frontier further.



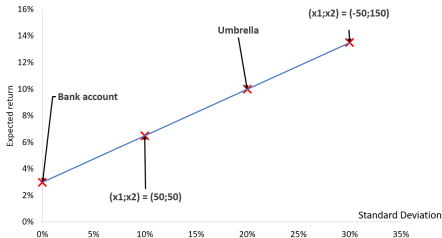
Recap: from two stocks to N

- ▶ A third stock turns the curve into a **region** of possible portfolios. The efficient frontier is its **upper-left edge**.
- ▶ Every stock we add, even a poor one on its own, moves the frontier further left: more diversification. The best frontier uses all available investments.



Recap: adding the risk-free asset

- ▶ The risk-free rate r_f is known in advance: zero volatility, zero correlation with every stock.
- ▶ A fraction x in a risky portfolio and $1 - x$ at r_f gives $E[R_p] = r_f + x(E[R_U] - r_f)$ and $\sigma_p = x\sigma_U$: both grow with x , so every mix is on a **straight line** from the risk-free point. Umbrella (10%, 20%) and a 3% deposit, half and half: 6.5% and 10%.



Recap: borrowing and leverage

- ▶ **Leverage** is investing with borrowed money. In the formulas it is a **negative weight on the risk-free asset**, $x_f < 0$, with more than 100% in the risky one.
- ▶ Borrow 50 at 3% and put 150 in Umbrella: $x_U = 1.5$, $x_f = -0.5$. Return 13.5% instead of 10%, volatility 30% instead of 20%: leverage scales the risk premium and the risk by the **same factor**.
- ▶ It takes you beyond the risky portfolio on the same straight line. Gains and losses are both magnified, so **never judge a return without its risk**.

Recap: the Sharpe ratio

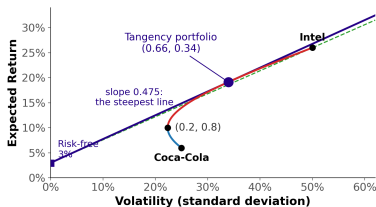
- ▶ The slope of the line from the risk-free point is the **Sharpe ratio**: the extra return you get per unit of risk.

$$\text{Sharpe ratio} = \frac{E[R] - r_f}{\sigma}$$

Portfolio	Expected return	Volatility	Sharpe ratio
Coca-Cola (0, 1)	6%	25%	0.12
(0.2, 0.8)	10%	22.4%	0.31
(0.5, 0.5)	16%	28%	0.46
Intel (1, 0)	26%	50%	0.46

- ▶ A steeper line pays more at every level of risk, whoever you are. With leverage any return is possible, so **compare Sharpe ratios, not returns**.

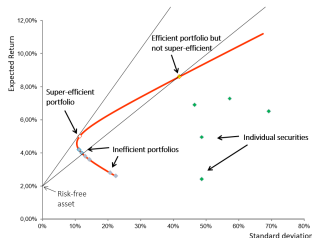
Recap: the Sharpe ratio and the efficient frontier



- The steepest line from the risk-free point is the one **tangent** to the curve. Once a risk-free asset exists, the efficient frontier **is that straight line**:

$$E(R_p) = r_f + \underbrace{\frac{E(R_T) - r_f}{\sigma_T}}_{\text{Sharpe ratio of } T} \times \sigma_p \quad \text{here } 3\% + 0.475 \times \sigma_p$$

Recap: the super-efficient portfolio

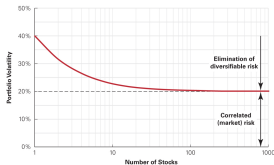


- ▶ The tangency point T is the risky portfolio with the **highest Sharpe ratio**: the tangency, or **super-efficient**, portfolio.
- ▶ **Everyone holds it.** Risk aversion only sets where you sit on the line: cautious investors lend part of their money at r_f , bold ones borrow and put more than their own money in T (two-fund separation).

Recap: two types of risk

Systematic risk: market-wide (economy, interest rates, inflation). It hits every stock at once, so diversification **cannot** remove it.

Idiosyncratic risk: specific to one firm (its products, its management). It averages out across many stocks, so diversification **does** remove it.



Volatility falls as stocks are added and flattens at the systematic level.

Beginning of lecture 4 content

Lecture 3 found the best portfolio of risky assets. Today we ask what that implies for the return each stock has to offer: the CAPM.

Introduction

- ▶ Today: how to determine the risk premium a stock must offer, with the **Capital Asset Pricing Model (CAPM)**: Tobin (1958), Treynor (1962), Sharpe (1964), Lintner (1965), Mossin (1966).
- ▶ **Why it matters.** The CAPM is the most widely used model linking risk and return. About three firms in four use it to estimate their **cost of capital**: the discount rate for projects, valuations and regulated prices.



James Tobin



William Sharpe



Harry Markowitz

Outline

1. Efficient portfolio and required returns (Chap. 11.6)
2. Capital Asset Pricing Model (Chap. 11.7)
3. Determining the risk premium (according to the CAPM) (Chap. 11.8)

A. Required returns: the answer first

- ▶ Last class we met the **Sharpe ratio**: the extra return we get for every unit of risk we take. Today we use it to decide whether an asset is **worth adding to a portfolio**.
- ▶ You hold a portfolio P . Someone offers you a stock A . **Should you buy some?**

The answer, in one sentence. A stock is worth adding only if its expected return pays for the risk it adds to *your* portfolio. And the risk it adds depends on how it **moves with what you already hold**, not on how risky it is on its own.

- ▶ The rest of this section unpacks that sentence in three steps:
 1. measure the risk A adds to P ;
 2. measure what that extra risk is worth;
 3. compare with what A actually pays.

Step 1: the risk a stock adds

- ▶ Add a small amount δ of A to P , financed by borrowing at r_f . The extra return is easy: $\delta(E(R_A) - r_f)$.
- ▶ The extra risk is not $\delta \sigma_A$. For a small δ the volatility of the portfolio rises by about

$$\Delta\sigma \approx \delta \times \sigma_A \times \rho_{A,P}$$

δ : the fraction of the portfolio put into A ; σ_A : the volatility of A on its own; $\rho_{A,P}$: the correlation between A and the portfolio P .

- ▶ The **correlation** is in the formula: a stock that moves with your portfolio adds its full risk, an uncorrelated one almost none, a negatively correlated one **reduces** risk. The same diversification story as Bore Industries.
- ▶ Where the formula comes from: the appendix at the end of the deck.

Step 2: what is that extra risk worth?

- ▶ You could take the same extra risk another way: **leverage**, borrowing to buy more of P .
- ▶ Leverage on P pays the Sharpe ratio of P for each unit of risk (Lecture 3). So extra risk $\Delta\sigma$ taken through P would earn

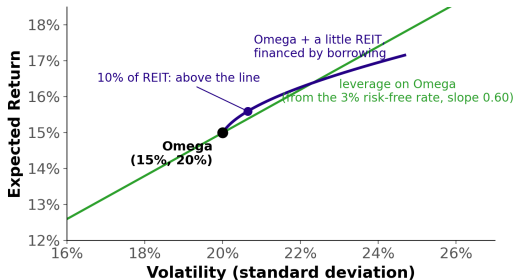
$$\Delta\sigma \times \frac{E(R_P) - r_f}{\sigma_P}$$

$\Delta\sigma$: the extra risk from step 1; $E(R_P) - r_f$: the risk premium of P ; σ_P : its volatility. The fraction is P 's Sharpe ratio.

- ▶ That is the **benchmark**. A is worth adding when its extra return (step 1, left) is at least what the same risk earns through P (right):

$$\delta(E(R_A) - r_f) \geq \Delta\sigma \times (E(R_P) - r_f) / \sigma_P$$

In a picture: Omega and the REIT



Zoom around Omega (15%, 20%). The green line is leverage on Omega. Adding a little REIT (9%, 35%, correlation 0.10) financed by borrowing traces the blue curve, which leaves Omega **above the line**: a small addition raises the Sharpe ratio. Add too much and the curve drops below it again.

Step 3: the rule, and the beta

- Buy A if its extra return beats the benchmark:

$$\delta (E(R_A) - r_f) \geq \delta \sigma_A \rho_{A,P} \times \frac{E(R_P) - r_f}{\sigma_P}$$

- Cancel δ and group the risk terms into one number:

$$\beta_{A,P} = \rho_{A,P} \frac{\sigma_A}{\sigma_P} = \frac{\text{Cov}(R_A, R_P)}{\text{Var}(R_P)}$$

The **beta** from Lecture 2, with P as the thing that moves:
when P moves by 1%, A moves by $\beta_{A,P}\%$ on average.

- The rule becomes a **required return** r_A : buy A if it pays at least r_A , short it if it pays less.

$$E(R_A) \geq r_A = r_f + \beta_{A,P} (E(R_P) - r_f)$$

Example: should we add the REIT to Omega?

- ▶ Omega: expected return 15%, volatility 20%. Treasury bills: 3%. REIT: expected return 9%, volatility 35%, correlation with Omega 0.10.
- ▶ Beta of the REIT with respect to Omega:
 $\beta = 0.10 \times 35\% / 20\% = 0.175$.
- ▶ Required return: $r_{\text{REIT}} = 3\% + 0.175 \times (15\% - 3\%) = 5.1\%$.
- ▶ The REIT pays 9%, above 5.1%: **yes, buy some**. It looks poor on its own (less return than Omega for more risk), but its low correlation means it adds very little risk.
- ▶ **The twist.** With a correlation of 0.50 instead of 0.10:
 $\beta = 0.875$, required return $3\% + 0.875 \times 12\% = 13.5\%$,
above the 9% it pays: **no**. Same stock, same return, same risk;
the correlation alone flips the decision.

From the required return to the CAPM

- ▶ The rule holds for **any** portfolio P : the required return of a stock is the risk-free rate plus its beta with respect to P times P 's risk premium.
- ▶ Lecture 3 showed that every investor holds the same risky portfolio, the tangency portfolio. Section B asks what that portfolio must be when everyone holds it, and finds the answer: **the market**.
- ▶ Replace P by the market portfolio and the required return becomes the **CAPM**.

B. Capital Asset Pricing Model

- ▶ Reminder: Markowitz (1952) + Tobin (1958) $\rightarrow$ any investor holds a combination of a risk-free investment and the super-efficient portfolio.
- ▶ How can we extend this result to a theory of market equilibrium? $\Rightarrow$ Let us add a couple of hypotheses to the previous result.

CAPM: hypotheses

- ▶ Starting point: Markowitz (1952) / Tobin (1958).

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- ▶ Investors hold homogeneous beliefs about expected returns, volatilities, and correlations of all financial assets.
- ▶ The market is in equilibrium.
- ▶ All investors hold an efficient portfolio, that is a portfolio which offers the highest expected return for a given level of volatility.

⇒ The super-efficient portfolio is unique.

Supply, demand, and the efficiency of the market portfolio

- ▶ Reminder: homogeneous beliefs $\rightarrow$ the investors all invest in the same super-efficient portfolio.
 - ▶ All risky assets are held by all investors in the same proportions (within the risky portfolio).
- ▶ But investors put different weights on the super-efficient portfolio and the risk-free asset depending on their risk aversion.
- ▶ Does this permit to pin down the composition of the super-efficient portfolio?

Supply, demand and the efficiency of the market portfolio: example

- ▶ Camille has identified the efficient portfolio and decides to hold it.
- ▶ She holds 10 000 EUR in stocks of Capgemini and 5 000 EUR in stocks of Sanofi.
- ▶ Tom, more risk averse, only holds 2 000 EUR invested in Sanofi stocks.
 1. Assuming that his portfolio is also efficient, what is his investment in Capgemini stocks?
 2. If all investors hold the efficient portfolio, what can we say about the total market capitalisation of Capgemini relative to the one of Sanofi?

The efficiency of the market portfolio

- ▶ It is easy to generalize the preceding example to any market.
 - ▶ Investors all hold the same efficient portfolio of risky assets (same weights).
 - ▶ Equilibrium between supply and demand → efficient portfolio contains all risky assets, weighted by their market capitalisation.
- ▶ **In equilibrium, the market portfolio is the super-efficient portfolio/tangency portfolio.**

The Capital Market Line

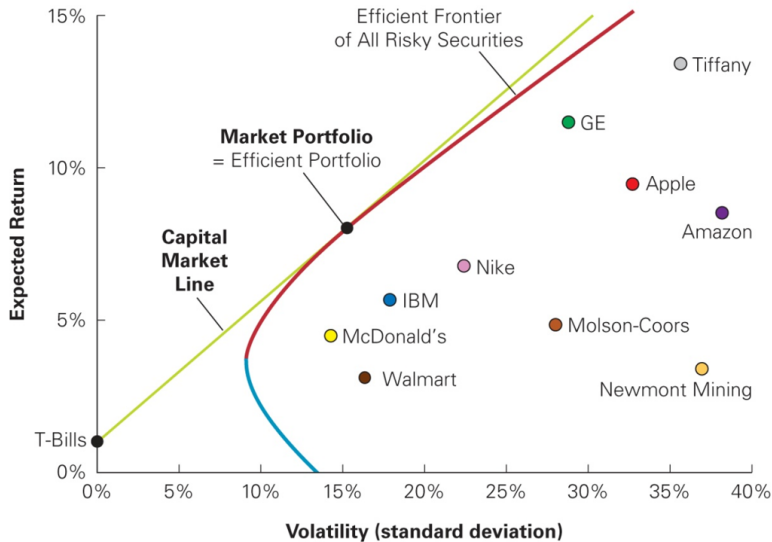
- ▶ The equation of the efficient frontier is then given by:

$$E(R_p) = r_f + \frac{E(R_M) - r_f}{\sigma_M} \sigma_P.$$

⇒ Equation of the Capital Market Line (CML)

- ▶ All portfolios on the CML are made up of the market portfolio and the risk-free asset!

Capital Market Line: example



C. Determining the risk premium

- ▶ Reminder: we are still trying to pin down an equilibrium relation for the returns on individual assets.
- ▶ Up to now, we know:
 - ▶ The required return on an asset (defined relative to any existing portfolio).
 - ▶ The composition of the portfolios on the efficient frontier (CML) + the composition of the portfolios held by each investor.
- ▶ It is thus straightforward to establish the relation between the required returns of an asset and the different market parameters!

Expected returns, required returns and market portfolio

- ▶ Special feature of any efficient portfolio: the expected return of all assets is equal to the required return (otherwise ...)!
- ▶ And yet, each investor holds the market portfolio! Thus, in equilibrium:

$$E(R_i) = r_i = r_f + \underbrace{\beta_i \times [E(R_m) - r_f]}_{\text{Risk premium of security } i} .$$

Market risk and beta

- ▶ The previous equation shows the “beta” of an asset with respect to the market portfolio:

$$\beta_i = \frac{\overbrace{\sigma_{R_i} \times Corr(R_i, R_m)}^{\substack{\text{Volatility of asset } i \text{ that} \\ \text{is common with the market}}} = \frac{Cov(R_i, R_m)}{Var(R_m)}.$$

- ▶ The required return of an asset depends on the sensitivity of that asset to the market risk.
 - ▶ It is thus independent from the idiosyncratic risk!

Example 1: expected return on a stock

- ▶ The risk-free rate is 4%.
 - ▶ The market portfolio has an expected return of 10% and a volatility of 16%.
 - ▶ Stock A has a volatility of 16% and a correlation with the market portfolio of 33%.
1. What is the beta of stock A and what is its expected return?
 2. What is the portfolio on the CML which has the same risk as Stock A and what is its expected return?

Example 2: expected return with negative beta

- ▶ The beta of the stock B is -0.3.
- ▶ The risk-free rate is 4% and the expected return on the market portfolio is 10%.

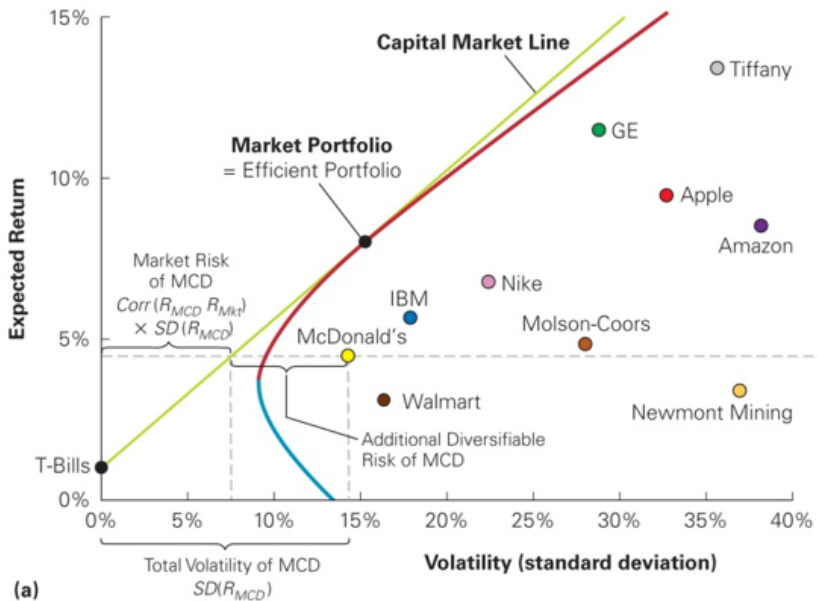
What is the expected return according to the CAPM?

What do you think about that?

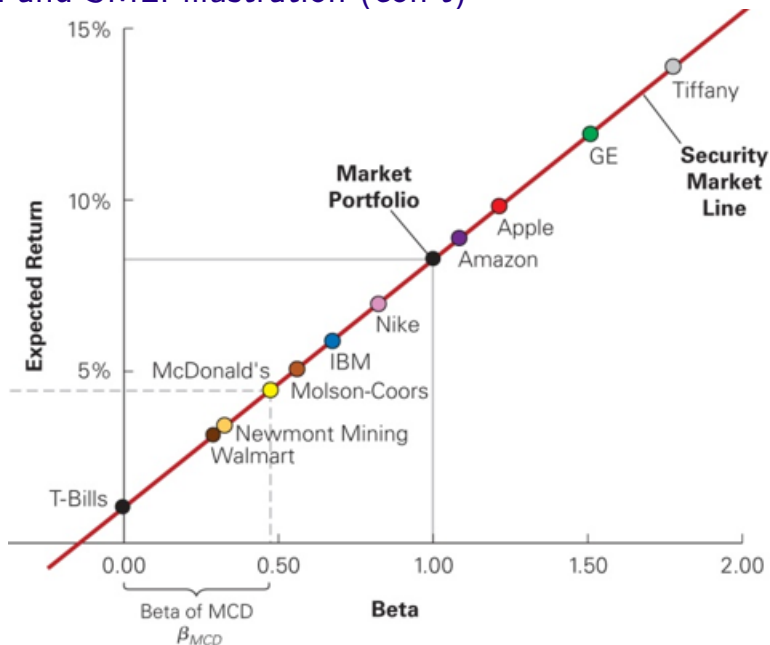
Security Market Line

- ▶ The CAPM highlights a linear relation between the beta of a security and the expected return.
- ▶ This straight line—called the Security Market Line—passes through the risk-free asset ($\beta = 0$) and the market portfolio ($\beta = 1$).
- ▶ In equilibrium, all assets are on the SML.
 - ▶ But no asset is on the CML!!

CML and SML: illustration



CML and SML: illustration (con't)



The beta of a portfolio

- The beta of a portfolio with return $R_p = \sum_i x_i R_i$ is given by:

$$\begin{aligned}\beta_p &= \frac{Cov(R_p, R_m)}{Var(R_m)} = \frac{Cov(\sum_i x_i R_i, R_m)}{Var(R_m)} \\ &= \sum_i x_i \frac{Cov(R_i, R_m)}{Var(R_m)} = \sum_i x_i \beta_i.\end{aligned}$$

- The beta of a portfolio is equal to the weighted average of the betas of its constituents.

Example: expected return of a portfolio

- ▶ The betas of Danone (BN.PA) and Ubisoft (UBI.PA) are respectively 0.5 and 1.8.
- ▶ The risk-free rate is 4%.
- ▶ The expected return on the market portfolio is 10%.

According to the CAPM, what is the expected return of an equally-weighted portfolio of these two stocks?

Summary: what to remember from the CAPM?

- ▶ The market portfolio is the only efficient portfolio. Hence, the highest expected return for a given level of volatility is achieved by combining the market portfolio with the risk-free asset (borrowing or lending).
- ▶ The risk premium of any asset is proportional to its beta with respect to the market. There is a linear relation between the expected return of an asset and its (systematic) risk called the SML.
- ▶ Remark: the hypotheses of the CAPM are quite restrictive. Nevertheless the CAPM is used widely to estimate the expected returns of assets or the cost of capital of projects.

Appendix: portfolio volatility approximation

In Section 1 of the slides, I claim that the approximation below holds:

$$\begin{aligned}\Delta\sigma &= \sqrt{\sigma_P^2 + \delta^2\sigma_A^2 + 2\delta\sigma_P\sigma_A\rho_{A,P}} - \sigma_P \\ &\approx \delta \frac{\sigma_P\sigma_A\rho_{A,P}}{\sigma_P}.\end{aligned}$$

To prove this result, define the function f as

$$f(\delta) := \sqrt{\sigma_P^2 + \delta^2\sigma_A^2 + 2\delta\sigma_P\sigma_A\rho_{A,P}}.$$

From Taylor's theorem, it follows that for δ close to zero, we have

$$f(\delta) \approx f(0) + f'(0)\delta.$$

Appendix: portfolio volatility approximation (con't)

Moreover, it is easy to check that

$$f'(\delta) = \frac{\delta\sigma_A^2 + \sigma_P\sigma_A\rho_{A,P}}{\sqrt{\sigma_P^2 + \delta^2\sigma_A^2 + 2\delta\sigma_P\sigma_A\rho_{A,P}}},$$

and, at $\delta = 0$, f and its first derivative simplify to $f(0) = \sigma_P$ and $f'(0) = \frac{\sigma_P\sigma_A\rho_{A,P}}{\sigma_P}$. Thus, Taylor's theorem implies that

$$\sqrt{\sigma_P^2 + \delta^2\sigma_A^2 + 2\delta\sigma_P\sigma_A\rho_{A,P}} \approx \sigma_P + \frac{\sigma_P\sigma_A\rho_{A,P}}{\sigma_P}\delta.$$

The approximation on the previous slide then follows almost immediately.